

FlashAttention Accelerator on PYNQ-Z2

Hardware Design — NYU Spring 2026

Anuj Apte ■ Ricardo Díaz ■ Raquel Brown

New York University

1. **Motivation**

The memory bottleneck in attention

2. **Algorithm**

Online softmax, tiled computation

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Interfaces & parameters

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7 sub-modules, HLS design

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Achievements & next steps

The Memory Bottleneck in Standard Attention

Standard scaled dot-product attention:

$$O = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

The $N \times N$ score matrix QK^T must be:

- Fully materialized before softmax
- Written to memory, then read back
- $4N^2$ bytes (float32)

For $N=512$: score matrix $\approx$ **1 MB**.

PYNQ-Z2 has only 140 BRAMs — it doesn't fit.

Our approach: FlashAttention

Process Q , K , V in small tiles.

Maintain running softmax state *on-chip*.

Peak memory: $O(N \cdot d)$ not $O(N^2)$

This project

- Vitis HLS implementation
- Target: **PYNQ-Z2** (Zynq XC7Z020)
- 4-channel AXI4 + AXI4-Lite control
- Verified against float64 golden model

Standard Attention — Numerical Example ($N=3$, $d=2$)

$$Q = K = V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad s_{ij} = \frac{q_i \cdot k_j}{\sqrt{2}}$$

Full 3×3 score matrix (all 9 values in memory)

	k_1	k_2	k_3
q_1	0.71	0.00	0.71
q_2	0.00	0.71	0.71
q_3	0.71	0.71	1.41

Softmax row q_1 :

$$e^{[0.71, 0.00, 0.71] - 0.71} = [1.00, 0.49, 1.00]$$

$$\ell = 2.49 \Rightarrow p_1 = [0.40, 0.20, 0.40]$$

$$o_1 = p_1 \cdot V = [0.80, 0.60]$$

All $N^2=9$ scores in memory at once

	k_1	k_2	k_3	
q_1	0.71	0.00	0.71	processing
q_2	0.00	0.71	0.71	
q_3	0.71	0.71	1.41	
	current row (q_1) other rows also loaded			

Softmax needs all 9 scores before it can compute a single probability.

FlashAttention — Same Example, $B_Q=B_K=2$

Focus on row $q_1=[1, 0]$. Init: $m=-\infty$, $\ell=0$, $o=[0, 0]$.

K/V tile 0 (k_1, k_2 on-chip)

- Scores: $[0.71, 0.00]$
- $m \leftarrow 0.71$, $e^{s-m}=[1.00, 0.49]$
- $\ell \leftarrow 1.49$
- $o \leftarrow \frac{1.00 \cdot [1, 0] + 0.49 \cdot [0, 1]}{1.49} = [0.67, 0.33]$

K/V tile 1 (k_3 on-chip)

- Score: $[0.71]$ (m unchanged $\Rightarrow$ correction $c=1$)
- $\ell \leftarrow 1 \cdot 1.49 + 1.00 = 2.49$
- $o \leftarrow \frac{1 \cdot 1.49 \cdot [0.67, 0.33] + 1.00 \cdot [1, 1]}{2.49} = [0.80, 0.60]$ ✓ same answer

Max on-chip at once: $1 \times 2 = 2$ scores.

Full 3×3 matrix never materializes.

3×3 score space Q tile 0 (q_1, q_2)

	k_1	k_2	k_3
	Pass 1		Pass 2
q_1	0.71	0.00	0.71
q_2			
q_3			

next Q tile

Numbers shown are q_1 's scores per tile.
Each pass loads only one K/V tile at a time.

Online Softmax: Exact Attention Without the Full Matrix

State per query row i (lives on-chip):

- m_i — running row maximum
- l_i — running normalization sum
- o_i — running output accumulator

Per K/V tile update:

$$m_i^{\text{new}} = \max(m_i, \max_j s_{ij})$$

$$l_i^{\text{new}} = \underbrace{e^{m_i - m_i^{\text{new}}}}_{\text{correction } c} l_i + \sum_j e^{s_{ij} - m_i^{\text{new}}}$$

$$o_i^{\text{new}} = \frac{c l_i o_i + \sum_j e^{s_{ij} - m_i^{\text{new}}} v_j}{l_i^{\text{new}}}$$

**o_i stays normalized after every tile.
No final division needed.**

Algorithm (BQ = BK = 8)

```
for q0 in 0, BQ, 2×BQ, ..., N:
    load Q tile                                (8 rows)
    init  $m = -\infty$ ,  $l = 0$ ,  $o = 0$ 
    for k0 in 0, BK, 2×BK, ..., N:
        load K, V tiles
        scores =  $Q_t K_t^T / \sqrt{d}$ 
        apply causal mask if needed
        update  $m$ ,  $l$ ,  $o$  online
    write output tile
```

- Correction $c = e^{m_i - m_i^{\text{new}}}$ rescales prior state when max grows
- Causal mask: $s_{ij} = -10^{30}$ for $j > i$, so $e^{s_{ij}} \rightarrow 0$ with no extra branch

IP Specification

Compile-time limits

N_MAX	64	max seq length
D_MAX	32	max dimension
BQ	8	Q tile size
BK	8	K/V tile size

Runtime (AXI4-Lite regs)

N	sequence length
d	embedding dim
causal	0 or 1
Q/K/V/O addr	64-bit DDR ptr

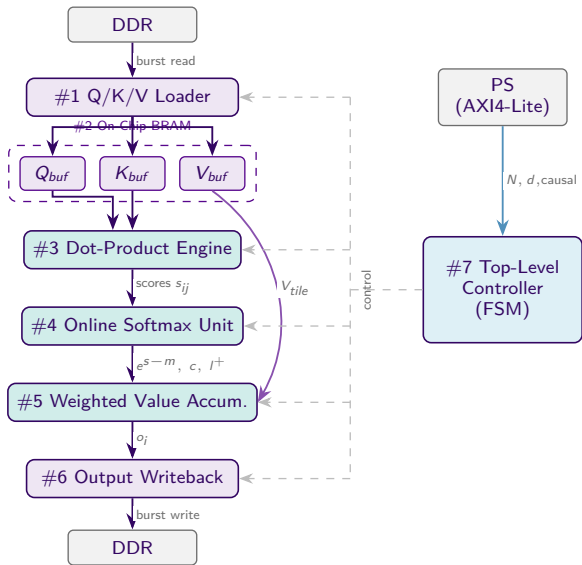
Hardware / Software boundary

- **PS (ARM):** allocates DDR buffers; writes parameters to AXI4-Lite registers; asserts AP_START; polls AP_DONE
- **PL (this IP):** reads tiles via AXI4 Master, computes on-chip, writes O back to DDR
- Intermediate state (m , l , scores, acc) *never* leaves the chip

AXI4 Master channels (no DDR contention)

gmem0	Q	read	32-bit / 16-burst
gmem1	K	read	32-bit / 16-burst
gmem2	V	read	32-bit / 16-burst
gmem3	O	write	32-bit / 16-burst

Architecture Overview



HLS Implementation Highlights

Dot-product engine: $II = 1$ via full unroll

```
SCORE_LOOP_J:
for (int j = 0; j < k_lim; j++) {
#pragma HLS PIPELINE II=1
    data_t dot = 0.0f;
    for (int x = 0; x < D_MAX; x++) {
#pragma HLS UNROLL // 32 parallel MACs
        if (x < d) dot += Qbuf[i][x]*Kbuf[j][x];
    }
    float s = dot * inv_sqrt_d;
    // causal: exp(-1e30) -> 0, no branch in softmax
    if (causal && (k0+j) > (q0+i)) s = -1e30f;
    scores[i][j] = s;
}
```

BRAM partitioned for 4-wide parallel reads

```
#pragma HLS ARRAY_PARTITION \
    variable=Qbuf cyclic factor=4 dim
    =2
#pragma HLS ARRAY_PARTITION \
    variable=Kbuf cyclic factor=4 dim
    =2
```

All inner loops: $II = 1$

Loop	II	Latency
LOAD_Q	1	12 cyc
LOAD_KV	1	12 cyc
SCORE_LOOP	1	202 cyc
FIND_MAX	1	3 cyc
UPDATE_ACC	1	34 cyc
STORE_O	1	9 cyc

Interface notes

- float *Q (1D) for AXI4 Master 2D arrays unsynthesizable
- int causal not bool 32-bit AXI4-Lite register
- hls::expf/sqrtf synthesizable math

Testbench pipeline

1. Python golden model generates vectors in **float64**
2. CSV loaded by C++ testbench (1D arrays, D_MAX stride)
3. `flash_attention_hls()` called in C-sim *and* RTL co-simulation identically
4. Max absolute error reported per test

Coverage

- Single-tile ($N \leq 8$)
- Multi-tile ($N=16, N=32$)
- **Partial tile** ($N=13, q_lim = k_lim = 5$)
- Max size ($N=64, d=32$)
- Full and causal masking
- Verified in Vitis HLS (C-sim & RTL co-sim)

Functional Verification — Results

Per-test results (Vitis HLS C-sim & RTL co-sim,
tolerance = 10^{-4})

ID	N	d	Causal	Max err vs float64	
0	4	4	No	6.0×10^{-8}	✓
1	4	4	Yes	1.2×10^{-7}	✓
2	8	4	Yes	1.2×10^{-7}	✓
3	16	8	No	2.1×10^{-7}	✓
4	16	8	Yes	1.2×10^{-7}	✓
5	13	8	No	3.6×10^{-7}	✓
6	13	8	Yes	2.4×10^{-7}	✓
7	32	16	No	1.8×10^{-7}	✓
8	64	32	No	3.0×10^{-7}	✓

9 / 9 PASS

Max error 3.6×10^{-7}

270× below 10^{-4} tol.

C-sim and RTL co-sim
results are identical.

Test 5 (N=13) is the worst case
partial tile boundary where the last
Q and K tiles are only 5 rows wide.

Synthesis Results — PYNQ-Z2 (XC7Z020, 100 MHz)

Resource Utilization

Resource	Used	Avail	%
BRAM_18K	141	280	50%
DSP48E1	178	220	81%
Flip-Flops	31 527	106 400	30%
LUTs	40 259	53 200	76%

Key driver: 32 fully-unrolled MACs consume **141 of 178 DSPs**.

Switching to `ap_fixed<16,6>` would cut DSP usage significantly.

Timing

Metric	Value
Target clock	10 ns (100 MHz)
Timing slack	-1.40 ns
Critical path	FIND_MAX chain
Est. max freq	≈ 87 MHz

AXI4 burst inference

- All 4 M_AXI ports: **burst inferred**
- Variable-length, up to tile size
- 4 separate DDR channels = no contention

Conclusions & Future Work

Delivered

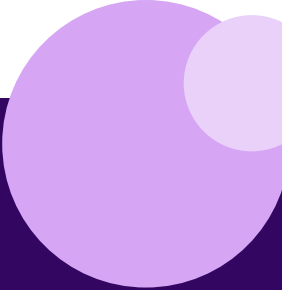
- Full FlashAttention forward pass, Vitis HLS
- PYNQ-Z2 (XC7Z020) — PS-ready IP block
- 4-channel AXI4 Master + AXI4-Lite control
- All inner loops: **II = 1**
- **9/9 tests pass** in Vitis HLS (C-sim & RTL co-sim)
- Max error: 3.6×10^{-7} (270× below tolerance)
- Optional causal masking at no latency cost

Limitations

- Timing violation (−1.40 ns at 100 MHz)
- DSPs at 81% limit D_MAX to 32
- Single head; sequential tile execution

Next steps

- Registered FIND_MAX tree → fix timing
- #pragma HLS DATAFLOW → overlap load / compute / store
- ap_fixed<16,6> → cut DSPs, quantify accuracy tradeoff
- Multi-head via time-multiplexing



Thank you

Questions?

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github.com/ricardotk002/hwdesign-flash-attn-accel